Example 7.20 The ratio of the volumes of two cones is 2:3. Find the ratio of their radii if the height of second cone is double the height of the first.

Solution Let r_1 and h_1 be the radius and height of the cone-I and let r_2 and h_2 be the radius and height of the cone-II.

Given that, $h_2 = 2h_1$ and $\dfrac{\text{Volume of the cone I}}{\text{Volume of the cone II}} = \dfrac{2}{3}$

$$\frac{\frac{1}{3}\pi r_1^2 h_1}{\frac{1}{3}\pi r_2^2 h_2} = \frac{2}{3} \Rightarrow \frac{r_1^2}{r_2^2} \times \frac{h_1}{2h_1} = \frac{2}{3}$$

$$\frac{r_1^2}{r_2^2} = \frac{4}{3} \Rightarrow \frac{r_1}{r_2} = \frac{2}{\sqrt{3}}$$

Therefore, ratio of their radii $= 2 : \sqrt{3}$

Progress Check

1. Volume of a cone is the product of its base area and _____.
2. If the radius of the cone is doubled, the new volume will be _____ times the original volume.
3. Consider the cones given in Fig.7.29
   (i) Without doing any calculation, find out whose volume is greater?
   (ii) Verify whether the cone with greater volume has greater surface area.
   (iii) Volume of cone A : Volume of cone B = ?

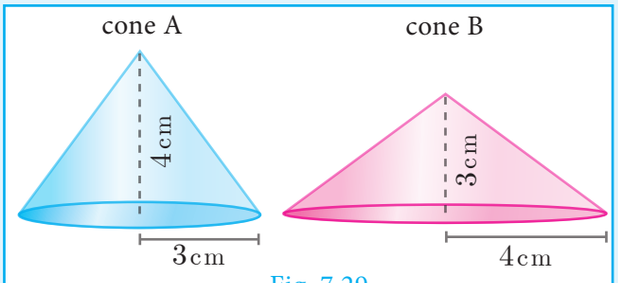
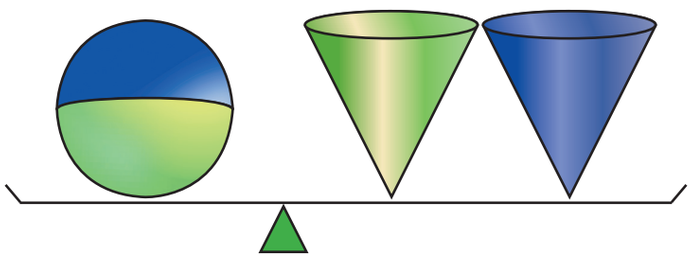


Fig. 7.29

7.3.4 Volume of sphere

Let r be the radius of a sphere then its volume is given by $V = \dfrac{4}{3}\pi r^3$ cu. units.

Demonstration



- Consider a sphere and two right circular cones of same base radius and height such that twice the radius of the sphere is equal to the height of the cones.
- Then we can observe that the contents of two cones will exactly occupy the sphere.

Fig. 7.30

10th_Maths_Chapter 7_English.indd 286 12/3/2024 2:59:41 PM